Problem 07: Electoral Boundaries

Time limit: 1 second

The changing dynamic of All Community Housing Society forces an early re-election for the president position. The society is very large and consists of many different blocks uniquely identified by a number ***1*** to ***N***. Some blocks are connected with each other and some not making them isolated. The election commission of the society received various suggestion for setting up polling booths across the society. The society wants to facilitate all the residents of the blocks to cast their votes at their ease.

The election commission wants to divide all these blocks into ***M*** distinct polling districts. However, there's a special requirement for the way these districts are organized:

- Every block must be assigned to exactly one polling district.
- For any two blocks that are directly connected by a road, the polling districts they belong to must be consecutive. In other words, if one block is placed in district ***X*** and its connected neighbor is in district ***Y***, then the absolute difference between ***X*** and ***Y*** must be exactly ***1***.

The challenge for the election commission is to determine the maximum number of polling districts ***(M)*** that can be created to facilitate voters. If it turns out to be impossible to arrange the blocks into such districts, the answer should be ***-1***.

Input

The first line of the input consists of ***t, (1 ≤ t ≤ 50)*** representing the total number of test cases given. The first line of the test cases represents, ***'N' (2 ≤ N ≤ 10^3)*** representing ***'N'*** blocks. The next line contains ***W (1 ≤ W ≤ 10^6)*** representing number of connecting roads. The next W lines contains two space separated block numbers representing a road between the two blocks.

Output

Output consists of t lines, each line contains exactly one number, representing the maximum number of distinct polling districts ***(M)***.

Sample input	**Sample Output**
4 6 6 1 2 1 4 1 5 2 6 2 3 4 6 3 3 1 2 2 3 3 1 3 1 2 3 6 5 1 2 2 4 1 5 4 3 3 6	4 -1 3 6